

Concept Note 1: Enhancing Disease Surveillance through Syndromic Data Analysis

Background

Early detection of disease outbreaks in Nigeria remains a challenge, especially in rural communities. The health system relies heavily on test results from labs, and that often takes time. By the time confirmed cases come in, the disease may have already spread.

Tracking symptoms like cough, fever, or diarrhoea before lab results can help spot potential outbreaks early. Nigeria already has systems like IDSR (Integrated Disease Surveillance and Response) in place, so this idea isn't starting from scratch. What's needed is to build on what's there and include more symptom-level data.

Objective

To improve disease outbreak detection by including syndromic data from local health centres and community reports into the existing disease surveillance system.

Implementation Strategy

Data Sources

- We'll work with systems already in use, especially DHIS2(District Health Information Software 2), which captures health facility reports. Other sources will include:
- Routine IDSR forms
- Reports from community health workers
- Mobile tools like ODK(Open Data Kit) and CommCare(Community Care), which are familiar to many fields staff

Data Processing

Health staff at the facility and community level will note common symptoms using simple forms. These will be submitted every week through the usual platforms.

At the LGA level, the team will clean and review the data. If numbers for any symptom shoot up beyond normal, it will be flagged for attention.

Data Analysis

We'll use basic analytics within DHIS2 to watch for unusual increases in cases. Once an alert is triggered, DSNOs(Disease Surveillance and Notification Officer) and health teams will be informed to carry out field checks.

- They'll respond within two days, especially if there's a risk of disease spreading.
- We'll test the system in a few LGAs first. This helps us learn and fix any issues before scaling.
- Getting local communities involved will make reporting more accurate and build trust. Training sessions will happen regularly to keep staff up to speed.

Conclusion

This idea builds on platforms Nigeria already uses, so it's not about starting over. By collecting simple symptom reports and responding faster, health officials can get ahead of outbreaks. With steady support, this model can be rolled out widely, maintained over time and improved upon.

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Concept Note 2: Developing a Machine Learning Platform to Address Health Inequalities

Background

There are health inequalities in many countries and communities. These differences are often due to income, gender, location, education, and access to a proper healthcare. As a result, some locations face more health problems than others. Most countries use health information systems like DHIS2(District Health Information Software 2), IDSR(Integrated Disease Surveillance and Response), and electronic medical records. However, these systems are mostly designed for data management not designed to run machine learning algorithms.

Machine learning platform can help governments and health organizations to identify, predict and Act on these inequalities.

Objective

To create and implement a machine learning-based digital platform designed to identify, visualize, and predict health inequalities. This platform will empower stakeholders to make proactive, equity-focused health decisions.

Implementation Strategy

Data Sources

We will integrate the below sources, transform them into a unified format, analyses, train, predict and diagnose/forecast.

Data Source	Description	Purpose	Frequency
Routine Health Data	From national health information systems like DHIS2, IDSR, and open-source electronic health records (EHRs)	Track service delivery trends and population coverage	Daily
Survey and Census Data	Leverage DHS (Demographic and Health Surveys) and national censuses	Provide demographic and socioeconomic input	Every 3-5 Years
Community and Mobile Data	From platforms like ODK, CommCare, and CHW applications	Reflect on-the-ground realities in real-time	Weekly
Geospatial and Environmental Data	Satellite imagery, facility locations, and climate variables	Understand factors influencing disease and access	Continuous or Periodic
Social Determinants	Data on education, income levels, housing, and access to water/sanitation	Capture broader factors driving health equity	Annual or Biennial

Data Processing

Data Source	Process 2 (Project team/Data Scientist)	Process 2(Data Analyst)	Process 3(Data Analyst)	Process 4(Data Analyst)
Routine Health Data	Collect data from DHIS2, EHRs, and IDSR	Clean and correct errors	Group data by region and time	Compare data with standards
Survey and Census Data	Collect survey and census data	Standardize the data	Fix inconsistencies	Combine with other relevant data
Community and Mobile Data	Gather real-time data from mobile apps	Sync data from devices	Check data for accuracy	Combine mobile data with other sources
Geospatial and Environmental Data	Collect satellite and environmental data	Process geospatial data	Merge environmental data with health data	Analyze data for health patterns
Social Determinants	Collect data on education, income, etc.	Clean the data	Standardize the data	Combine with health data for analysis

Data Analysis

Use of Machine learning technologies including **Supervised Learning** (Regression & Classification Models), Time Series Analysis, Anomaly Detection: etc with focus would on Predicting disease outbreaks, Monitor service delivery trends, forecast healthcare needs and identify underserved areas or population and create a monthly location-based reporting with recommendation management decisions making.

Conclusion

Health inequality can be managed and resolved with the necessary strategic approach, technology; in this case machine learning can change the narrative especially for underserved locations. The efficiency of the process and ethical standards would be managed by the quality assurance team.

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